

AI Adoption Control Plane for AWS CloudWatch

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Executive Summary

Enterprise AI adoption has hit an inflection point: 72% of organizations have agentic AI in production, but a 60% governance gap persists. Every major observability vendor is racing to become the 'AI control plane.' ServiceNow launched AI Control Tower (May 2026) with 5 pillars and 30+ integrations. Dynatrace positioned its platform as 'the control plane for AI in production.' Datadog shipped LLM Observability with agent monitoring. AWS has Bedrock Guardrails + CloudWatch metrics but no unified AI governance experience. The opportunity: a CloudWatch-native AI adoption control plane that unifies Bedrock guardrail monitoring, LLM observability, cost tracking, and governance into a single console.

Competitor Landscape

ServiceNow AI Control Tower

Current state (May 2026): Five-pillar platform - Discover, Observe, Govern, Secure, Measure. 30+ enterprise integrations (AWS, GCP, Azure, SAP, Oracle, Workday). Acquired Traceloop for deep AI agent observability and Veza for identity security. Kill switches for rogue agents.

Evolution: Basic AI inventory (late 2024) -> governance layer (Q1 2025) -> full 5-pillar platform with enforcement (Knowledge 2026, May).

Strengths: Broadest enterprise integration footprint (30+). Cross-platform discovery. Enforcement muscle (real-time kill switches). Token cost tracking across OpenAI, Anthropic, Google.

Weaknesses: ITSM-centric DNA. Premium pricing. Not cloud-native - sits on top of cloud providers rather than inside them.

Datadog LLM Observability

Current state: GA product covering LLM monitoring, agentic AI monitoring, LLM experiments, AI Agents Console. Auto-instrumentation for OpenAI, LangChain, Bedrock, Anthropic. Execution flow visualization.

Strengths: Developer-first UX. Best-in-class trace visualization. Token usage and cost per tool/workflow branch. Sensitive data scanning and prompt injection detection.

Weaknesses: No governance/enforcement layer. No enterprise-wide AI discovery. No business outcome measurement.

Dynatrace

Current state: Intelligence Platform with unified observability for major agentic frameworks (Bedrock AgentCore, Google ADK, OpenAI Agent, Anthropic MCP, LangChain). MCP Server for AI assistants. AI-powered forecasting.

Strengths: Deterministic + agentic AI fusion. Model versioning with A/B testing. ServiceNow partnership for remediation. Strongest topology mapping.

Weaknesses: Expensive. Complex deployment. ServiceNow dependency. Not developer-friendly.

Splunk (Cisco)

Current state: AI Agent Monitoring (GA Q1 2026), AI Infrastructure Monitoring. GPU metrics, tokenomics (time-to-first-token, costs), quality metrics (hallucinations, bias, drift, accuracy).

Strengths: OpenTelemetry-native. NoSample tracing. Infrastructure-layer depth. Cisco network integration.

Weaknesses: Query complexity barrier. AI governance secondary. No business-outcome layer.

AWS Current State

Current state: Bedrock Guardrails (6 safeguard policies), CloudWatch metrics (invocation latency, tokens, guardrail enforcement), CloudTrail API logs. Cross-account guardrails available.

The Gap: Everything exists in pieces but there is no 'AI Control Plane' experience that ties guardrails + metrics + cost + agent monitoring into one view. This is the product opportunity.

Quantitative Data

Metric	Value	Source	Date
Enterprises with agentic AI in production	72%	Agentic AI Institute	2026
Governance gap	60%	Agentic AI Institute	2026
Organizations facing AI adoption challenges	79%	Writer survey	2026
CIOs: unchecked AI is serious concern	76%	Logicalis CIO Report	2026
Companies with mature AI governance	12%	Cisco Privacy Benchmark	2026
CIOs compromising governance	62%	Logicalis	2026
C-suite: AI 'tearing company apart'	54%	Writer	2026
AI observability market 2025	\$1.1B	Precedence Research	2025
AI observability projected 2035	\$3.29B	Precedence Research	2025
AI in observability growth	+\$2.91B, 22.5% CAGR	Technavio	2025
ServiceNow AI Control Tower integrations	30+	Knowledge 2026	May 2026
Bedrock Guardrails block rate	88%	AWS	2026

Gaps: What Nobody Does Well Yet

- 1. Unified AWS-native AI governance** - Bedrock Guardrails exist but no CloudWatch console ties guardrails + metrics + cost + agent monitoring into one view.
- 2. Business outcome correlation** - Everyone tracks tokens and latency, nobody ties AI investment to business metrics (conversion, CSAT, resolution time).
- 3. Multi-model cost optimization** - Cost tracking exists but nobody offers 'switch from GPT-4 to Claude Haiku and save 40% with <5% quality degradation' recommendations.
- 4. AI adoption maturity scoring** - ServiceNow does inventory/discovery but nobody provides a maturity model scoring organizational AI health.
- 5. Cross-account AI governance** - Bedrock has cross-account guardrails but no cross-account AI visibility dashboard for AWS Organizations.

Key Takeaways for PRD

- 1. The gap is real and urgent:** 72% have AI in production but 60% lack governance. AWS has the pieces but no unified experience.
- 2. ServiceNow is the primary competitor** with 5-pillar AI Control Tower. But they are ITSM-centric and sit ON TOP of cloud. AWS wins by being INSIDE the cloud.
- 3. 'AI Control Plane' is the positioning** - Dynatrace already uses this language. AWS should own it for cloud-native.
- 4. Business outcome correlation is the white space** - Nobody connects AI spend to business results. This is the differentiated value prop.
- 5. Must support non-Bedrock AI** - A Bedrock-only solution is a non-starter. Customers use OpenAI, Anthropic, self-hosted alongside Bedrock.
- 6. Prescriptive > configurable** - 62% of CIOs lack governance knowledge. The product must guide users, not just provide dashboards.
- 7. Cross-account governance is the AWS-native moat** - No competitor can do this as deeply as CloudWatch.

Sources

1. Agentic AI Institute - Agentic AI Enterprise Adoption 2026 (72% production, 60% governance gap)
2. ServiceNow Newsroom - AI Control Tower expansion, Knowledge 2026 (5-pillar, 30+ integrations)
3. Datadog - LLM Observability product page (agent monitoring, execution flow)
4. Futurum Group - Dynatrace Perform 2026 analysis (AI control plane positioning)
5. Splunk - AI Agent Monitoring innovations blog (GA Q1 2026)
6. Writer - Enterprise AI Adoption 2026 survey (79% face challenges)
7. Logicalis - CIO Report 2026 (76% unchecked AI concern, 62% compromising governance)
8. CIO.com / Cisco - Privacy Benchmark 2026 (12% mature governance)
9. Precedence Research - AI Observability Market (\$1.1B 2025)
10. Technavio - AI in Observability Market (+\$2.91B, 22.5% CAGR)
11. AWS - Bedrock Guardrails documentation (6 policies, 88% block rate)
12. Deloitte - State of AI in the Enterprise 2026